

Job Portal Platform

Master Integration Specification

A merged, comprehensive architecture blueprint mapping data flows, data model payloads, systemic logic, and integration test scenarios.

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Integration Interface Topology

The diagram below tracks the lifecycle of an integration event as it passes securely between your frontend presentation layer, middleware schemas, and persistent infrastructure.



1. Core Integration Module Specifications

1.1 Authentication Module

- **How It's Done:** The platform decouples user identity management from transactional application logic. The React frontend captures client login actions and passes them down to the Django REST authorization layer. Upon verification against the security gateway, a local JSON Web Token (JWT) is generated.
- **Session Tracking:** The frontend Axios interceptor automatically catches this secure string and attaches it to the standard HTTP header (Authorization: Bearer <Token_Value>) for all subsequent server-side network requests.

Authentication Payload Schema

FRONTEND INTERFACE PARAMETER	BACKEND MODEL TARGET	TECHNICAL FORMAT	VALIDATION CONSTRAINT RULES
registrationEmail	email	String (RFC 5322)	Required, unique check against database, max 255 chars.
registrationPassword	password	String (Hashed)	Minimum 8 characters, complex character sets required.
userRoleType	role	String Enum	Strict validation matches: ['SEEKER' , 'EMPLOYER' , 'ADMIN'].

1.2 Employer Subscription & Workspace Boundaries

- **How It's Done:** Subscriptions act as automated validation gates before the system accepts data updates. Whenever an employer creates a listing, a backend filter checks the organization's tier rules stored in the database.
- **Tier Limits:**
 - *Starter Plan (Free Tier):* Restricted to exactly 20 published positions. Analytics tracking fields and candidate advanced queries are disabled.
 - *Premium Plan (Tier Value: ₹799/mo):* Grants extended validation boundaries up to 50 active positions and unblocks analytics tracking screens.
 - *Enterprise Plan (Tier Value: ₹2000/mo):* Complete system allocation bypass. Unlimited active postings, dedicated technical profile mapping, and immediate direct messaging.

Subscription Payload Mapping

FRONTEND DATA ORIGIN	BACKEND RELATIONAL FIELD	DATA DEFINITION	TRANSACTIONAL LOGIC EXPECTED
planSelectionId	tier_id	UUID / Int	Verifies identity configuration state against pricing engine parameters.
paymentValue	amount_billed	Decimal (10,2)	Must map exactly to chosen layout config, ledgered in currency format.
validityPeriod	expiration_timestamp	DateTime ISO	Sets programmatic expiration timestamp (Current Date + Selected Term).
transactionStatus	billing_state	String Enum	Strict status flow handling: ['PENDING' , 'SETTLED' , 'DECLINED'].

1.3 Job Management Lifecycle

- **How It's Done:** The platform's job creation form matches your backend fields one-to-one. The frontend packs user inputs (such as job titles, industrial lookups, and mode selections like Hybrid, Remote, or On-site) into a structured payload.
- **Data Protection:** The Django framework captures this object, sweeps it through an HTML script-sanitizer to block unwanted scripts, and creates a clean record inside the system database tables.

Data Payload Synchronization Matrix

FRONTEND ENTRY ELEMENT	BACKEND PERSISTENT FIELD	TYPE DEF	DATA INTEGRITY RULE
Job Title Input	title	VarChar(255)	Required field; e.g., "Fullstack Developer". Strip script tags.
Industrial Type Selector	industry_segment	VarChar(100)	Matches pre-approved system lookup categories.
Work Type Radio Selection	work_mode	Enum String	Strictly enforces configuration boundaries: ['Hybrid', 'Remote', 'On-site'].
Salary Field (Max CTC)	compensation_lpa	Decimal(4,2)	Numeric parsing up to 99.99 LPA. Optional field.
Experience Range Value	experience_months	Integer Range	Converts raw user ranges into standard localized base-month integers.
Job Description Text Area	description_body	Text Block	Sanitized Markdown. Content-length enforced (min 50 chars).

1.4 Application Tracking & File Storage

- **How It's Done:** Applications process text data and physical files at the same time. Candidate details travel as standard structured data, while raw resume binaries are uploaded through a dedicated multipart/form-data pathway.
- **File Handling:** The backend saves file binaries into protected file directories and links a clean URL string reference to the user's application row. State changes follow a strict sequential ledger: ['Applied', 'Reviewing', 'ShortListed', 'InterviewScheduled', 'Rejected'].

Application Event Transmission Schema

```
{
  "target_job_id": 4092,
  "candidate_profile_id": 9912,
  "applicant_legal_name": "Harshavardhan Reddy",
  "contact_email_verified": "applicant@domain.in",
  "resume_attachment_reference": "/storage/binaries/resumes/res_962a.pdf",
  "application_initial_state": "Applied"
}
```

1.5 Real-Time Communication

- **How It's Done:** Chat messaging runs on automated background updates. When an employer or candidate sends a message, the platform matches it to the conversation group using validation rules.
- **Smart Fallbacks:** If network quality drops, the application automatically falls back to optimized REST API loops to log text data securely along with server-side UTC timestamps.

Chat Data Mapping

DATA POINT ATTRIBUTE	DATABASE ASSIGNMENT	STRUCTURAL RESPONSIBILITY
sender_identity_token	sender_id_fk	Identifies the originating session owner profile.
receiver_identity_token	receiver_id_fk	Maps data boundary directly to target user dashboard.
message_payload_string	message_body	Stores rich message text content, stripping executable scripts.
temporal_epoch	created_at_dt	Appends server-validated UTC epoch timestamps upon ingestion.

1.6 Event Notification Relay

- **How It's Done:** System notifications rely on an internal trigger engine. When a critical action finishes (like moving an applicant to "Shortlisted" or creating a billing invoice), an active background worker takes over.
- **Alert Channels:** This worker updates individual in-app notification count badges in real time while pushing automated transactional emails directly through secure outbound mail protocol gateways.

Notification Channel Actions

SYSTEM TRIGGER EVENT	PRIMARY IN-APP CHANNEL ACTION	ASYNCHRONOUS OUTBOUND SMTP PAYLOAD
Application Shortlisting Event	Pushes active banner data update to candidate navigation panel.	Transmits complete interview schedule options and venue links.
Subscription Invoice Issuance	Updates billing ledger view rows instantly.	Generates and emails a compliant PDF receipt record.
New Applicant Submission	Increments employer overview counters in real time.	Sends an daily summary report detailing new matching profile cards.

2. Configuration & Integration Validation

2.1 Technical Environment Settings

To keep deployment environments safely separated, all global configuration values are managed in standalone system environment files:

```
# Base Application Access Routing
BASE_GATEWAY_URL=https://api.jobportal.enterprise.internal/v1
PERSISTENCE_DB_ENGINE=django.db.backends.mysql

# Transactional Outbound Delivery Gateways
OUTBOUND_SMTP_HOST=smtp.internal-relay.net
OUTBOUND_SMTP_PORT=587
SECURITY_TLS_ENFORCE=True

# Multi-Tenant Structural Allocation Limits
MAX_FREE_TIER_POSTINGS=20
MAX_PREMIUM_TIER_POSTINGS=50
```

2.2 Integration Test Matrix

The reliability of your operational boundaries can be validated across system borders using these clear test verification criteria:

TEST REF	MODULE TARGET	INTEGRATION TRIGGER CONDITION	EXPECTED SYSTEM RESULT
TC-AUTH-01	Authentication	Client submits authentication payload containing an external gateway token string.	System constructs localized internal session and grants view access permissions.
TC-SUBS-02	Billing Engine	Profile tries to post a 21st job listing while restricted to the Free Starter tier.	Database updates are blocked; interface handles a 403 Entitlement Exhausted status code.
TC-JOB-03	Job Lifecycle	Form data block is transmitted through the job deployment entry route.	Engine filters scripts, returns a 201 Created code, and logs entries into the repository.
TC-CHAT-04	Communication	User sends a text message payload within an active dialogue conversation.	Message is pushed to active web listeners and committed to background history tables.

2.3 Platform Version Roadmap Plan

This integration framework scales systematically across predictable deployment iterations:

- **Version 1.0 (Current Deployment):** Stabilizes identity security layers, baseline recruitment data definitions, transactional multi-tiered billing parameters, binary file attachments, and linear dashboard reports.
- **Version 2.0 (Planned Scaling Horizon):** Introduces parsing engines for uploaded file data blocks, algorithmic candidate routing matching scores, video interview scheduling parameters, and real-time dashboard tracking visualizations.